

SFFS-PC-NN optimized by genetic algorithm for dynamic prediction of financial distress with longitudinal data streams

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ABSTRACT

Recently, research of financial distress prediction has become increasingly urgent. However, existing static models for financial distress prediction are not able to adapt to the situation that the sample data flows constantly with the lapse of time. Financial distress prediction with static models does not meet the demand of the dynamic nature of business operations. This article explores the theoretical and empirical research of dynamic modeling on financial distress prediction with longitudinal data streams from the view of individual enterprise. Based on enterprise's longitudinal data streams, dynamic financial distress prediction model is constructed by integrating financial indicator selection by using sequential floating forward selection method, dynamic evaluation of enterprise's financial situation by using principal component analysis at each longitudinal time point, and dynamic prediction of financial distress by using back-propagation neural network optimized by genetic algorithm. This model's ex-ante prediction efficiently combines its ex-post evaluation. In empirical study, three listed companies' half-year longitudinal data streams are used as the sample set. Results of dynamic financial distress prediction show that the longitudinal and dynamic model of enterprise's financial distress prediction is more effective and feasible than static model.

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1. Introduction

Research of enterprise's financial distress prediction (FDP) can provide guidance for enterprise's financial risk precaution, because it can generate early warning signals before the outbreak of financial crisis. Thus, it can effectively reduce the possibility of firms being trapped in financial difficulties due to the further deterioration of financial situation. Theoretical and empirical research of dynamic modeling on FDP with longitudinal data streams from the view of individual enterprise has important practical and theoretical research value.

Faced with the challenges of economic globalization, the traditional financial theory about continuous operation faces more criticism. Existing static models for FDP are not able to adapt to the situation that the sample data flows constantly with the lapse of time. FDP with static models does not meet the demand of the dynamic nature of business operations. Financial distress definitions from different researches are of difference. Carminchael [3] presented that financial difficulty is a situation that an enterprise encounters frustration when it tries to fulfill its obligations, which includes insufficiency of liquidity, insufficiency of equity, default of

debt, and insufficiency of liquid capital. Doumpos and Zopounidis [11] proposed that financial difficulties contain negative net asset value, inability to repay creditors' debt, overdraft of bank deposits, inability to pay preferred stock dividends, deferred payment, deferred due interest and principal payments, among others. Ross et al. [18] summarized previous studies and concluded that financial difficulties consist of the following four conditions: (1) business failure, that is, the company cannot pay the outstanding debt after liquidation; (2) legal bankruptcy, namely: the company or creditors apply to the court for a declaration of bankruptcy; (3) technical bankruptcy, namely: the company cannot fulfill the contract on scheduled to repay principal payment and interest; and (4) accounting bankruptcy, namely: the company's book net assets are negative. Chen [6] proposed three characteristics of financial distress, which are substantial weakening of corporate profitability, weakening of solvency and lost or approximate lost of business continuous operations.

When an enterprise has not fallen into the "absolute" state of financial distress yet, the enterprise's managers need to compare the current financial situation with the historical financial positions to analyze the enterprise's future trend, aiming at enhancing the capacity to withstand financial risk and improving the level of financial control. The definition of an enterprise's relative deterioration of financial situation always changes with the process of its

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life cycle, which is called longitudinal concept drift of enterprise financial distress in this research. With the lapse of time, financial situations of an enterprise on different particular time points have different definitions, which is consistent with the law of enterprise's development. For instance, financial indicators of an enterprise at certain year may be normal, even showing significant improvement of financial situation in comparison with those of previous years. However, in a new rapidly changing environment, the financial indicators of the enterprise at that year may indicate that the enterprise was staying in a poor financial situation in comparison with its current financial indicators. That is a so-called relative financial distress. Thus, along with the advance of enterprise life cycle, definitions of financial distress should be dynamically changed. Consequently, longitudinal concept drift of relative enterprise financial distress is introduced in this article, which is defined as the situation that profitability, solvency and capability of continuous operation are relatively deteriorating in comparison with the neighbor historical financial situations.

2. Literature review

Theoretical and empirical researches of financial distress prediction have been widely focused on. Scholars have tried their best to explore methods for financial distress prediction in the last decades, and achieved fruitful results.

In theoretical studies, traditional FDP methods include: univariate discriminant analysis, multivariate discriminant analysis (MDA), logistic analysis, among others. Scholars are taking more interests in artificial intelligence technology, in which neural network is one of the most widely used promising tools [14,17,12,8]. For example, Demyanyk and Hasan [10] presented that Back-propagation neural network (BPNN) could overcome the classification restriction of single-layer network, which is one of the most commonly used methods in classification and prediction. In addition, other artificial intelligence methods have been tried in FDP, such as case-based reasoning [2,15,4], support vector machine [13,9], and so on.

In empirical studies, data sets for FDP are mainly panel-based, which are mostly collected by matching distress and non-distress enterprises at certain time points. In addition, these data sets are mainly for long-term static prediction modeling prior to more than 1 year [1,13,19]. Some examples are as follows. Lin [16] took financial data of Taiwan ST (specially treated) and non-ST paired companies from 1998 to 2005 as samples, and used MDA, logit, probit and artificial neural network (NN) models in financial distress prediction. Chen et al. [7] took 68 companies' financial data as initial data, and combined NN with data mining technology to build FDP model. This research provides a new means for this topic, namely: dynamic FDP based on a single enterprise's longitudinal financial data. In comparison with panel data's spatiality, longitudinal data puts more emphasis on time series of samples.

Though theoretic and empirical researches of financial distress prediction have made great achievements, yet there are several issues that need to be stressed or focused. (1) The current studies of FDP are still in static modeling stage. Sample data which is collected from a specific period is static. Thus, the demand of dynamic financial operation is not taken into account. Static models cannot meet the need of financial distress prediction in new economic environment and business operating environment with the lapse of time. There is an urgent need to study dynamic modeling of FDP based on data streams. (2) Most of existing studies in financial distress prediction take long-term prediction prior to more than 1 year, instead of short-term or medium-term prediction prior to no more than 1 year. They do not put enough attention on financial

distress prediction of an individual enterprise with longitudinal historical data. (3) Longitudinal FDP models should not forecast the current financial state/situation with the current published financial data. This type of discrimination and prediction belongs to ex-post evaluation. Thus, effectiveness of this type of models is unrealistically high, which results in the lack of persuasiveness.

Therefore, we construct a new method by combining sequential floating forward selection with principal component neural network (SFFS-PC-NN) and optimize it with genetic algorithm in order to dynamically forecast financial distress prior to 0.5 year. The medium-term longitudinal data stream is from enterprise's semi-annual reports. The financial data of period $(t - 0.5)$ are used as inputs to predict the financial situation of period t , and the outputs are compared with evaluated results from principal component analysis (PCA) based on financial data of the period t . Thus, we combine ex-ante prediction and ex-post evaluation to verify the model's effectiveness.

3. Dynamic modeling of FDP with longitudinal data streams

3.1. Dynamic modeling design

Assume that half a year is the period interval of financial indicators inflow. The width of the modeling sample is set as n , which means that, at each of the current time point, a training data set consisting of n instances based on recent periods' data streams is constructed for FDP modeling. The training data set contains n samples. The time point of the recent $(n + 1)$ period is denoted as follows:

$$t = t_0 + 0.5(i_0 - 1) \quad (i_0 = 1, 2, \dots, n, n + 1) \quad (1)$$

where t_0 denotes the first period time point in the recent $(n + 1)$ periods, i_0 denotes the serial number of every time point. For instance, $i_0 = (n + 1)$ accords with the time point $t^* = t_0 + 0.5n$. The initial feature set of financial indicators is expressed as $F' = \{f'_1, f'_2, \dots, f'_{m_0}\}$. Accordingly, initial data set for dynamic modeling of financial distress prediction which consists of the recent $(n + 1)$ periods' m_0 financial indicators is denoted as follows:

$$X^0 = \{x^0_{i_0 j_0}\} = \begin{bmatrix} x^0_{11} & x^0_{12} & \cdots & x^0_{1m_0} \\ x^0_{21} & x^0_{22} & \cdots & x^0_{2m_0} \\ \vdots & \vdots & \ddots & \vdots \\ x^0_{n1} & x^0_{n2} & \cdots & x^0_{nm_0} \\ x^0_{(n+1)1} & x^0_{(n+1)2} & \cdots & x^0_{(n+1)m_0} \end{bmatrix} \quad (2)$$

$(i_0 = 1, 2, \dots, n, n + 1; j_0 = 1, 2, \dots, m_0)$

Fig. 1 shows the framework of dynamic modeling for financial distress prediction with longitudinal data streams. The frame is composed of data preprocessing module, financial situation discrimination (FSD) module and FDP module.

In data preprocessing module, feature set for discrimination of financial situation is filtered out from initial feature set F' . This process can efficiently improve the model's accuracy. The filtered feature set is denoted as $F = \{f_1, f_2, \dots, f_m\}$. With the filtered feature set F , the corresponding financial feature data is extracted from initial data set X^0 . The extracted financial data is used as the input of FSD module and it is denoted as follows:

$$X^1 = \{x^1_{i_1 j_1}\} = \begin{bmatrix} x^1_{11} & x^1_{12} & \cdots & x^1_{1m} \\ x^1_{21} & x^1_{22} & \cdots & x^1_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x^1_{(n+1)1} & x^1_{(n+1)2} & \cdots & x^1_{(n+1)m} \end{bmatrix} \quad (3)$$

$(i_1 = 1, 2, \dots, n, n + 1; j_1 = 1, 2, \dots, m)$

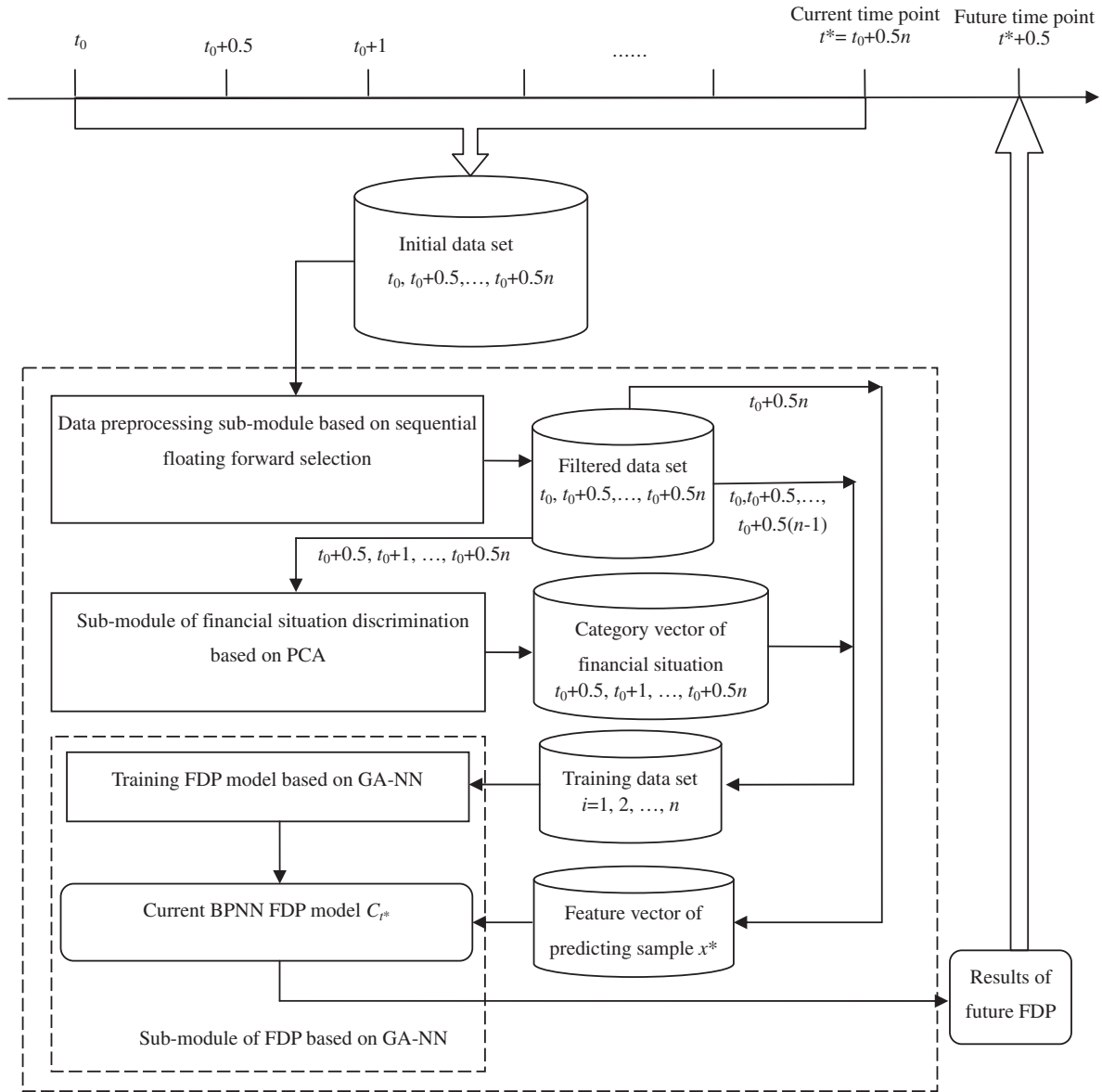


Fig. 1. Framework of dynamic FDP modeling with longitudinal data streams.

In FSD module, enterprise's relative financial situations at recent n time points are evaluated with PCA method. Compared with other evaluation methods, PCA can not only effectively and properly evaluate an enterprise's relative financial situation at every longitudinal time point, but also play a positive role in the further dimension reduction. The output vector of FSD module, which shows relative financial situation category of each period, is denoted as follows:

$$Y' = [y'_2, \dots, y'_n, y'_{(n+1)}]^T \quad (4)$$

FSD module only realizes the ex-post financial evaluation, which cannot meet the management requirements of ex-ante financial prediction. Therefore, FDP module is designed based on the FSD module to predict financial distress constantly and dynamically with longitudinal data streams. FDP module is combined with FSD module organically. Outputs of FSD module is taken as training target outputs of FDP module. Prediction result of FDP module is simultaneously compared with discrimination result of FSD module of the next period. By this means, we can evaluate

efficiency of enterprise's current financial activities, and effectively enhance the enterprise's capacity to withstand financial risks. Assume that the period of financial distress prediction is half a year, which means predicting financial situation half a year in advance. Use the feature data of the period $(t - 0.5)$ to predict financial situation category of the period t . The training data set is denoted as follows:

$$TD = \{(x_i, y_i)\} \quad (i = 1, 2, \dots, n) = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} & y_1 \\ x_{21} & x_{22} & \cdots & x_{2m} & y_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} & y_n \end{bmatrix} = \begin{bmatrix} x_{11}^1 & x_{12}^1 & \cdots & x_{1m}^1 & y'_2 \\ x_{21}^1 & x_{22}^1 & \cdots & x_{2m}^1 & y'_3 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1}^1 & x_{n2}^1 & \cdots & x_{nm}^1 & y'_{(n+1)} \end{bmatrix} \quad (5)$$

BPNN optimized by GA (GA-BPNN) has shown its strong macro-searching and good global optimization capability for financial distress prediction. At the current time point ($t^* = t_0 + 0.5n$), use TD (formula (5)) as the training sample data set to build an effective FDP model C_{t^*} . Put the feature data $x^* = (x_{(n+1)1}^1, x_{(n+1)2}^1, \dots, x_{(n+1)m}^1)$ at this time point ($t^* = t_0 + 0.5n$) into the model C_{t^*} to predict the financial situation at the future time point ($t^* + 0.5 = t_0 + 0.5(n+1)$).

With the lapse of time, financial data that reflects true financial situation at the time point ($t^* + 0.5 = t_0 + 0.5(n+1)$) will continue to flow in. Since the new data implies the new features of financial distress concept in the new phase, it is necessary to consider a new dynamic predictive modeling with the new data in order to continuously maintain the dynamic adaptability to the current concept features of financial distress.

3.2. Financial indicator selection with sequential floating forward selection method: input of NN

In order to remove noisy data, we construct the model of financial distress prediction by using sequential floating forward selection (SFFS) method. SFFS inserts the most significant features into the chosen feature subset from the remaining features at each time, and dynamically deletes the most meaningless features from the chosen feature subset. Feature selection for the optimal subset is a complex combinatorial optimization problem, and sub-optimal algorithm like SFFS is a suitable method to solve such problems [5]. SFFS can not only control repeated cross-validation to reduce the calculation burden, but also improve the correct classification rate and reduce the execution time by inserting and deleting steps [20].

Sequential forward selection (SFS) is used to initialize SFFS and the most distinguishing feature subset is selected by inserting and deleting repeatedly. First, the initial data set X^0 is normalized by the following formula.

$$X'_{j_0} = \frac{X_{j_0}^0 - A_n \times \min(X_{j_0}^0)}{\max(X_{j_0}^0) - \min(X_{j_0}^0)} \quad (6)$$

where A_n denotes an n -order matrix with all 1 elements. Initialize X^1 and m ($X^1 = \phi$, $m = 0$), and then select three features with SFS into X^1 to obtain the initial chosen feature subset, which is denoted as $X^1(3) = \{X_{j_1}^1\}$, $j_1 = 1, 2, 3$. Suppose that the cost function is $J(X^1(m))$, insert features by using the following mechanism.

$$X^+ = \arg \max_{X \in X^1 - X^1(m)} J(X^1(m) + X); \quad X^1(m+1) = X^1(m) + X^+, \quad m = m + 1 \quad (7)$$

Fourth, delete features by using the following mechanism.

$$X^- = \arg \max_{X \in X^1(m)} J(X^1(m) - X) \quad (8)$$

If $J(X^1(m) - \{X^-\}) > J(X^1(m))$, the identification capacity of the removed subset is enhanced, then delete X^- ;

$$X^1(m-1) = X^1(m) - X^-, \quad m = m - 1 \quad (9)$$

Otherwise, re-insert the feature. Repeat inserting and deleting features until m is equal to the predetermined number. The obtained result is expressed as $X^1(m)$.

3.3. Financial situation discrimination with principal component analysis: target output of NN training

PCA is a multivariate statistical method about multi-attribute decision making and comprehensive evaluation, which can correctly and effectively evaluate an enterprise's operations and financial situation with the process of reducing dimension of

multivariate data. PCA model is built with historical financial indicators. The model is used to discriminate an enterprise's financial situation at each period according to the enterprise's comprehensive financial evaluation scores. Since the evaluation scores are derived from the enterprise's own historical financial data at different periods, the results have longitudinal comparability, which reflect the relative financial distress in the process of business operations.

The basic idea of PCA is to replace original related financial indicators with new reformed ones that are unrelated. The process is as follows. First, import the data set

$$\begin{aligned} A &= X^1 - \{x_{j_1}^1\} \quad (j_1 = 1, 2, \dots, m) \\ &= \begin{bmatrix} x_{21}^1 & x_{22}^1 & \dots & x_{2m}^1 \\ x_{31}^1 & x_{32}^1 & \dots & x_{3m}^1 \\ \vdots & \vdots & \ddots & \vdots \\ x_{(n+1)1}^1 & x_{(n+1)2}^1 & \dots & x_{(n+1)m}^1 \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix} \\ &= \{a_{ij}\} \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, m) \end{aligned} \quad (10)$$

Let A^* be the new standardized data set, that is

$$A^* = \begin{bmatrix} a_{11}^* & a_{12}^* & \dots & a_{1m}^* \\ a_{21}^* & a_{22}^* & \dots & a_{2m}^* \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}^* & a_{n2}^* & \dots & a_{nm}^* \end{bmatrix} = \{a_{ij}^*\} \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, m) \quad (11)$$

in which a_{ij}^* is denoted as the follows:

$$a_{ij}^* = \frac{a_{ij} - \bar{a}_j}{\sqrt{\text{VAR}(\bar{a}_j)}} \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, m) \quad (12)$$

Second, test the correlation among the m financial indicators from a correlation matrix R_0 by calculating the formula (12).

$$r_{j_1 j_2} = \frac{\sum_{i=1}^n (a_{ij_1}^* - \bar{a}_{j_1}^*)(a_{ij_2}^* - \bar{a}_{j_2}^*)}{\sqrt{\sum_{i=1}^n (a_{ij_1}^* - \bar{a}_{j_1}^*)^2} \sqrt{\sum_{i=1}^n (a_{ij_2}^* - \bar{a}_{j_2}^*)^2}} \quad (13)$$

$(j_1 = 1, 2, \dots, m; j_2 = 1, 2, \dots, m)$

Third, calculate eigenvalues λ_j ($j = 1, 2, \dots, m$) and eigenvectors e_j ($j = 1, 2, \dots, m$) by Jacobi method. The Jacobi method is as follows. First, make a series of orthogonal rotation to matrix R_0 by matrices $B_1, B_2, \dots, B_k$. Thus, the symmetric matrix R_0 turns into diagonal matrix R_k , where $R_k = (B_1, B_2, \dots, B_k)^T R_0 (B_1, B_2, \dots, B_k) = B_k^T R_{k-1} B_k$ ($k = 1, 2, \dots$). It is an iteration process until convergence. Then, the m diagonal elements in diagonal matrix R_k are eigenvalues, while each column of the transformation matrix product is the corresponding eigenvectors. Order λ_j in accordance with their magnitudes, that is: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m \geq 0$. Then contribution rate of each principal component is calculated as follows:

$$\beta_j = \frac{\lambda_j}{\sum_{j=1}^m \lambda_{j_1}} \quad (j = 1, 2, \dots, m) \quad (14)$$

The cumulative contribution rate is calculated through the follows:

$$G(j) = \frac{\sum_{j_1=1}^j \lambda_{j_1}}{\sum_{j_1=1}^m \lambda_{j_1}} \quad (j = 1, 2, \dots, m) \quad (15)$$

Fourth, extract eigenvalues ($\lambda_1, \lambda_2, \dots, \lambda_p$) whose cumulative contribution rates are up to 85%, and they correspond to p ($p \leq m$) principal components by computing its component matrix as follows:

$$l_{kj} = \sqrt{\lambda_k e_{kj}} \quad (k = 1, 2, \dots, p; j = 1, 2, \dots, m) \quad (16)$$

$$\begin{cases} z_{11} = l_{11}a_{11}^* + l_{12}a_{12}^* + \dots + l_{1m}a_{1m}^* \\ z_{12} = l_{21}a_{11}^* + l_{22}a_{12}^* + \dots + l_{2m}a_{1m}^* \\ \dots\dots\dots \\ z_{ip} = l_{p1}a_{11}^* + l_{p2}a_{12}^* + \dots + l_{pm}a_{1m}^* \end{cases} \quad (17)$$

$(i = 1, 2, \dots, n)$

$$Z = \begin{bmatrix} z_{11} & z_{12} & \dots & z_{1p} \\ z_{21} & z_{22} & \dots & z_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n1} & z_{n2} & \dots & z_{np} \end{bmatrix} \quad (18)$$

Fifth, the enterprise's financial comprehensive evaluation function is as follows:

$$S = \beta_1 Z_1 + \beta_2 Z_2 + \dots + \beta_p Z_p \quad (19)$$

Higher comprehensive score indicates better financial situation at that period. Finally, determine the years whose scores are in the top 55% as financial health, which is denoted as $y_i = (1, 0)$. The remains are in financial distress, which is denoted as $y_i = (0, 1)$.

3.4. GA-NN module

BP neural network is a one-way transmission multilayer feed-forward network. For a neuron, the network input can be denoted as the follows:

$$net = X_1 w_1 + X_2 w_2 + \dots + X_m w_m \quad (20)$$

where $X_1, X_2, \dots, X_m$ denote network inputs and $w_1, w_2, \dots, w_m$ denote corresponding neural connection weights. The output of the neuron is expressed as the follows:

$$y = f(net) = \frac{1}{1 + e^{-net}} \quad (21)$$

Transfer function of BP neural network must be differentiable. Its division of region consists of nonlinear hyper-plane, which is a

relatively smooth surface. Thus, it has better accuracy and fault tolerance than linear classifications. The structure of BP neural network, initial connection weights, and threshold selection are of great importance to train a network. Genetic algorithm can be used to optimize the neural network.

Genetic algorithm is a computational model derived from Darwin's organic evolution process of genetic selection and natural elimination, which forms a kind of search algorithm with the process of "generating + testing". Through a comprehensive simulation of natural selection and genetic mechanism, it has strong macro-searching and good global optimization capability. The optimal neural network weights and thresholds are searched in training process by genetic algorithm. The combination of genetic algorithm and BP neural network accomplish a widely mapping capability of neural network. With the global searching capability of genetic algorithms, NN can not only effectively avoid the local minimum problem, but also speed up the learning of network.

Calculation process of neural network model optimized by genetic algorithm is shown in Fig. 2, and each chromosome in the genetic initial population consists of BP neural network's weights and thresholds. Through genetic selection, crossover and mutation, the weights and thresholds are assigned as W_{ih} , W_{ho} , B_h , and B_o in the final chromosome. Based on the global optimization algorithm, quantified conjugate gradient method of BP algorithm has efficient local searching capability which can further calculate the network weights and thresholds. Specific process is listed as follows:

First, initialize parameters, including: learning rate η , maximum learning frequency M , and presumptive learning error ε . Second, import an enterprise's m financial indicators of n periods as the input training samples, and import results of determinations by PCA as the target training outputs. Training data set is denoted as $\{X, Y\}$. Third, calculate outputs of neurons in hidden layer and output layer by activation function; compute the error function between target output vector $Y = \{y_i\}$ ($i = 1, 2, \dots, n$) and network inference output vector $Y'' = \{y''_i\}$ ($i = 1, 2, \dots, n$). Fourth, modify the connection weights and thresholds by comparing E .

$$E = \frac{1}{2n} \sum_{i=1}^n \sum_{o=1}^2 (y_i(0) - y''_i(o))^2 \quad (22)$$

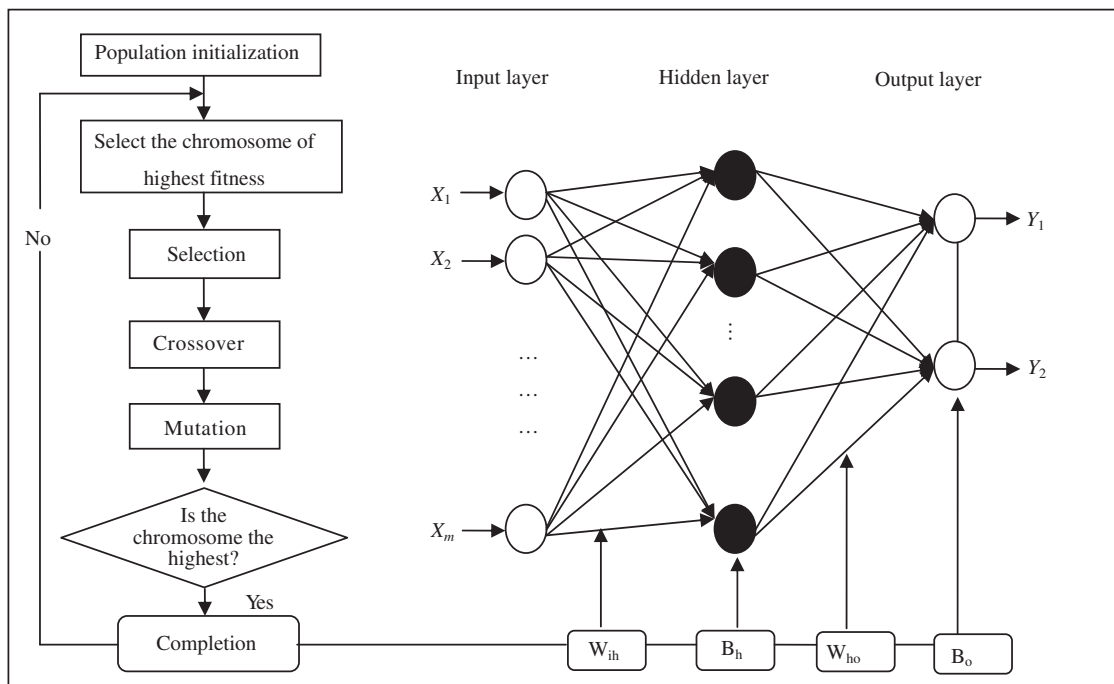


Fig. 2. GA optimized BP-NN model.

The process of training is effective when $E < \varepsilon$ or the learning frequency exceeds M . Finally, take the financial indicators of the time point $t^* = t_0 + 0.5n$ as testing sample. The output is $y''_{(n+1)}$, namely: the prediction result of the time point $(t^* + 0.5) = t_0 + 0.5(n + 1)$.

4. Data experiment

4.1. Experimental data collection

In the experiment, the samples should come from an enterprise which has a relatively long history as subject. We select a Chinese listed company which is established in 1992 and is still listed now from “Chinese Stock Listed Companies” to collect the samples. Its stock code is 600606. We chose a company from real estate industry for the following reasons. (1) Real estate industry is an important industry in the national economy of a country for its large finance amount, long industry chain, and wide economic impact. (2) Real estate industry has characteristics of large industry's profits and asset inflation, which attract a large number of funds involved. Lots of equity investors are interested in investing in this industry. If financial situation of companies in this industry can be well predicted, managers can improve the risk prevention awareness and grasp the development trend of the financial situation of enterprises. Meanwhile, regulators can stabilize the market, and investors can make the right investment decisions. (3) Compared with disclosed reports of Chinese listed companies from other industries in early years, companies in real estate industry posted more detailed information in statements. It is more convenient for data collection. (4) The model is concerned about the longitudinal historical data of individual enterprises, which needs a long-term historical data. In China, the history of real estate industry is long. Most companies in this industry survive for at least 15 years. Since public reports of the enterprise at the periods of 1992–1994 are not available, financial data of the company from mid-1995 to mid-2010 is used. If the predicted year is defined as t , the advanced half year is defined as $(t - 0.5)$.

According to the principles of comprehensiveness, generality, sensitivity, measurability and apriority, the candidate financial indicators for FDP are presented in Table 1, which includes 16 financial indicators from five areas of solvency, profitability, operational capacity, growth capacity and per share indicators. The Chinese Ministry of Finance requires publish of corporate cash flow statement since 1998. Thus, there is no published relevant data about cash flow statement before 1998. The candidate financial indicator system does not involve indicators directly related with cash flows. With the accumulation of financial data, further studies can include indicators related with cash flow in practical application.

The enterprise's financial indicator data is collected from its published financial reports every six months in the period of mid-1995 (1995m) to mid-2010 (2010m). It means that financial data of each period of January to June and each period of July to December is collected to construct the experimental data set, which contains 31 samples of the enterprise.

4.2. Experiment design

FDP modeling based on longitudinal data streams derives a model which dynamically and continuously updates to meet the current financial distress concept feature, and then updated model is used to predict the financial situation of the next period. In order to test validity of the proposed approach, the experiment simulates the process of time lapse to train and test FDP model constantly and dynamically. The process of experimental modeling and testing is designed as Fig. 3. Based on the enterprise's financial indicator data from 1995m to 2010m, 23 half-years are set as the

Table 1

Candidate financial indicators for financial distress dynamic prediction.

Type	No.	Indicators
Solvency	V1	Current ratio
	V2	Cash ratio
	V3	Liability-asset ratio
	V4	Rate for long-term assets
Profitability	V5	Return on equity
	V6	Return on assets
	V7	Main operating margins
Operational capacity	V8	Turnover ratio of receivable
	V9	Turnover of current assets
	V10	Turnover of fixed assets
	V11	Turnover of total capital
Growth capacity	V12	Growth rate of main business
	V13	Net profit growth rate
	V14	Expansion rate of investment in fixed assets
Per share indicators	V15	Net assets per share
	V16	Earnings per share

moving width. At each current time point in the process of time lapse, features are selected from the 16 candidate financial indicators by SFFS method. The company's financial situations are dynamically discriminated by PCA. Prediction modeling and testing are based on GA-BPNN.

Assume 2006m as the first time point of FDP model. Neural network model based on financial data of 2006m and the years before is used to predict the financial situation of 2006y. Import the financial indicator data from 1995y to 2006y into FSD module, and discriminate the financial situation at every half year point. Outputs of FSD module about the enterprise's financial situations from 1995y to 2006m are taken as the training target output labels in NN modeling. The discriminated financial situation of 2006y is taken as the testing target label in NN model testing. To achieve the effect of combining ex-ante prediction with ex-post evaluation, this experiment uses financial indicator data from 1995m to 2005y to discriminate financial situations from 1995y to 2006m, all of which are used to train the neural network model. Financial indicators data of 2006m is imported to the model to predict the financial situation of 2006y. Prediction accuracy is evaluated by comparing the output with the discriminative result of FSD module of 2006y.

With the lapse of time, we assume that the time point of FDP modeling shifts as the cycle of half a year from 2006m to 2010m. Throughout the experiment, original data set of each experiment also shifts as the cycle of half a year to maintain the same width, which means that at each time the longitudinal experimental data set removes the financial data of its first period, and adds the financial data of its most recent period to form dynamic longitudinal data streams. Thus, the experiment moves seven times to verify the model's stability and vertical generalized prediction ability. Finally, the financial indicators of 2010m is taken as the inputs of FDP model at the time point of 2010m, the financial situation at the time point of 2010y is to be predicted.

4.3. Parameters setting

The financial situation of enterprise is divided into two types: i.e., distress and non-distress/health. The former discrimination result is denoted as (0, 1), while the latter as (1, 0). Let 0.5 be the threshold. The definitions of 0 and 1 are with the variance of 0.5. For example, values in $[-0.5, 0.5)$ belong to category 0. Values in $[0.5, 1.5]$ belong to category 1. In genetic algorithm, the search scope is from -1 to 1 . The fitness function was set as follows:

$$RMSN = \sqrt{\sum_{i=1}^n \sum_{o=1}^2 (y_i(o) - y'_i(o))^2} \quad (23)$$

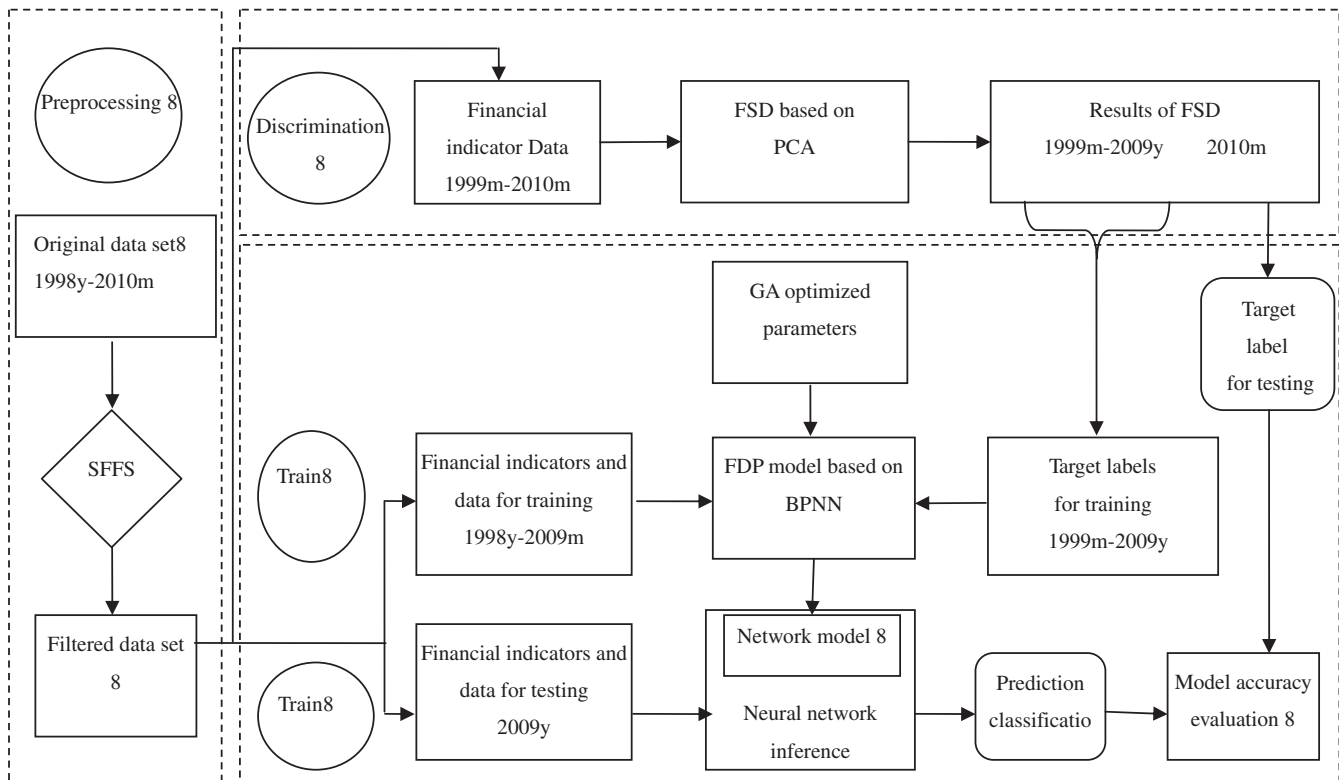
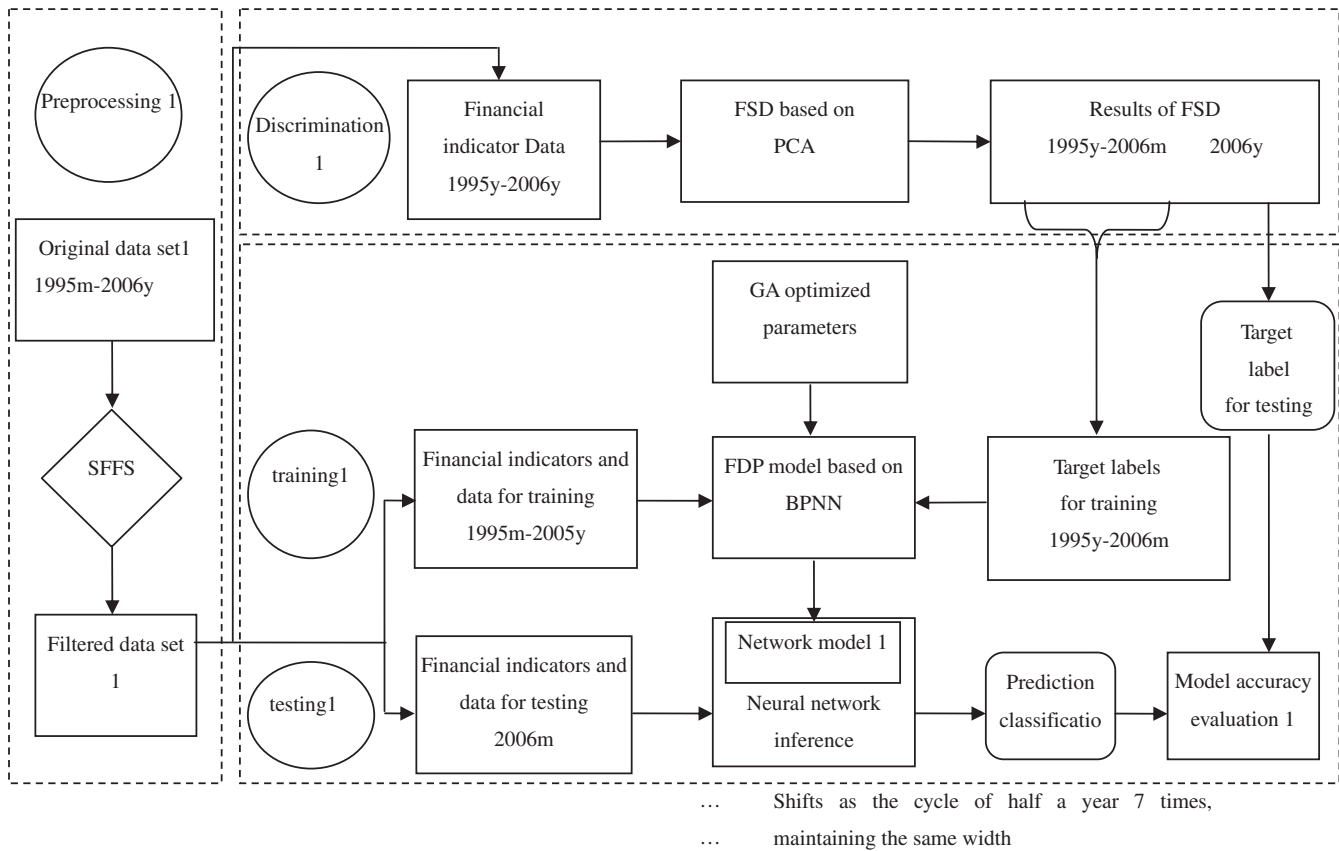


Fig. 3. Experimental design to test the model's validity.

BPNN model is a three-layer network, with which the number of neurons in input layer is set by the number of financial indicators after feature selection. The number of neurons in output layer is set as 2. Accordingly the number of neurons in the hidden layer is set as 10 by Kolmogorov theorem. Thus, the network structure is 8-10-2. The learning rate is set as 0.001, the maximum training time is set as 5000, and the target error is set as 0.00001. Activation function of hidden layer is tansig, while the output layer's activation function is purelin. The training function is the quantified conjugate gradient function of trainscg. Since the outputs are Boolean variables, the actual input variables need to be normalized. The normalized data is used to train the network where the initial weights and thresholds are assigned by genetic algorithm.

4.4. Experimental results and analysis

Taking the case of Model 1, we first use financial indicator data from 1995m to 2006y for data preprocessing. The selected financial indicator data by SFFS is shown in Table 2. This filtered data is input to FDP module based on PCA. Discriminations of financial situations are shown in Table 3. The training data set of NN modeling is constructed by paring the financial indicator data from 1995m to 2005y in Table 2 with the target output labels from 1995y to 2006m in Table 3. The initial population size of genetic algorithm is set as 50. It can be seen from Figs. 4 and 5 that after searching 100 generations, the error of genetic algorithm has minimized, and the average fitness of chromosomes has stabilized when it is up to 0.0781.

As shown in Fig. 6, the network stops training at the 221th step when the error is up to the target value. Finally financial indicator data of 2006m in Table 2 is input to the NN model to predict the financial situation of 2006y. The output is (1.0892, -0.0977), indicating that the financial situation of the company in 2006y is non-distress. This result is consistent with the result of principal component discrimination of (1, 0), which proves that the prediction of Model 1 is correct.

Similarly, remove the invalid and weakly identified features by SFFS to obtain the significant financial indicators in every moving

model, as shown in Table 4. Such preprocessing not only saves time, but also ensures the accuracy of prediction. Finally, the dynamic prediction is verified eight times. The network's testing categories are compared with the discrimination categories by FSD. The results are shown in Table 5. It can be seen that the network's prediction results are practically in accordance with the discriminated results, indicating that the model for FDP is appropriate. In addition, compared with the dynamic GANN model, existing best static GANN model has a wrong warning in these eight periods. Absolute errors between network inference outputs and target sample outputs in static model are larger, which shows dynamic model for financial distress prediction is more appropriate.

With the lapse of time, use the financial indicator data from 1999m to 2009y and the discriminated financial labels from 1999y to 2010m to train the ninth model, which is utilized to predict the company's financial situation of 2010y based on the financial indicator data of 2010m. Result of the network inference is (0.9988, 0.0037), meaning that the category is not financial distress. However, the company got entangled in financial trouble few years ago, from which managers should draw a lesson. Through the records of the causes, solutions and results of financial distress, the company can avoid the recurrence of similar financial distress.

In order to verify the horizontal general applicability of this model, the company with stock code of 600620, which belongs to real estate industry, and the company with stock code of 600613, which belongs to pharmaceutical manufacturing industry, was selected as general samples. Take the same time span, the same financial indicators and the same model parameters to verify the dynamic prediction eight times. As can be seen in Table 6, the company in the same industry has only one incorrect warning in the process; while the company in a different industry has three wrong warnings. Thus, a dynamic FDP model based on an individual company's longitudinal financial data is more suitable for FDP of the company itself, but its generalization applicability on enterprises in the same industry is acceptable and is better than that on firms from different industry. If the model is to be used to forecast financial distress of some other company, it is a good

Table 2
Values of the enterprise's financial indicators.

Year	Current ratio V1	Liability-asset ratio V3	Net assets per share V15	Rate for long-term assets V4	Cash Ratio V2	Turnover of fixed assets V10	Main operating margins V7	Return on assets V6
1995m	1.0556	1.4445	2.0700	1.5600	0.4054	1.1588	0.0424	0.0074
1995y	1.0803	1.4863	1.9000	1.4300	0.0481	2.0685	0.0600	-0.0103
1996m	1.0191	1.4398	1.6900	1.3300	0.4054	0.9446	-0.1698	-0.0371
1996y	0.9595	1.5229	1.7400	1.0600	0.0708	1.6455	0.0800	-0.0111
1997m	0.9570	1.4564	1.6200	1.0500	0.4054	0.8695	0.0190	0.0047
1997y	0.9991	1.4786	1.6300	1.1000	0.0637	1.8891	0.0239	0.0015
1998m	1.0145	1.6048	1.5400	1.1000	0.1016	0.9467	0.0877	-0.0220
1998y	1.1600	1.4251	1.8800	1.6700	0.1358	1.6729	0.2552	0.0631
1999m	1.2235	1.4699	1.9600	2.1400	0.1142	0.4400	0.4362	0.0124
1999y	1.2690	1.4901	2.3600	2.6400	0.1616	6.4501	0.2279	0.0514
2000m	1.1713	1.3969	1.3000	1.8700	0.2798	2.1216	0.3869	0.0104
2000y	1.1580	1.3635	1.4000	2.1100	0.3177	3.6353	0.1759	0.0354
2001m	1.1501	1.3123	1.4600	2.0600	0.2787	6.5008	0.2463	0.0168
2001y	1.1600	1.3200	1.7300	2.1300	0.3100	6.4508	0.1930	0.0415
2002m	1.9075	2.0710	4.2550	6.9400	0.8552	9.4771	0.2835	0.0128
2002y	1.5866	1.8639	3.7700	15.2900	0.6684	12.1645	0.0964	0.0189
2003m	1.6889	2.0463	3.8620	14.0600	0.6888	6.1886	0.3100	0.0116
2003y	1.6103	1.9064	4.0300	12.2300	0.5294	5.4359	0.1118	0.0200
2004m	1.4822	1.7331	3.9910	14.0800	0.4701	5.4252	0.2274	0.0058
2004y	1.5228	1.7359	4.2900	17.8700	0.4268	9.9486	0.1649	0.0266
2005m	1.6588	1.9040	4.3310	19.9200	0.3002	9.5999	0.4078	0.0058
2005y	1.5241	1.8207	4.1300	20.9200	0.3863	2.4026	-0.0331	0.0251
2006m	1.7178	2.0211	4.1780	23.0300	0.3042	1.7804	0.6271	0.0156
2006y	1.7291	1.9042	3.9800	3.2500	0.4156	7.1888	0.2594	0.0309

Table 3
Discriminated results of financial situations.

Year	Comprehensive scores of principal components	Sequence	Discrimination of financial situation	Target outputs
95y	0.6297	1	Health	(1, 0)
96m	−0.0066	17	Distress	(0, 1)
96y	0.0549	8	Health	(1, 0)
97m	0.0445	13	Distress	(0, 1)
97y	−0.0456	21	Distress	(0, 1)
98m	0.1064	4	Health	(1, 0)
98y	0.0794	5	Health	(1, 0)
99m	−0.0033	16	Distress	(0, 1)
99y	0.3737	3	Health	(1, 0)
00m	0.0123	14	Distress	(0, 1)
00y	−0.0675	22	Distress	(0, 1)
01m	0.0603	7	Health	(1, 0)
01y	−0.0425	20	Distress	(0, 1)
02m	0.0475	10	Health	(1, 0)
02y	0.0076	15	Distress	(0, 1)
03m	−0.0319	19	Distress	(0, 1)
03y	0.4103	2	Health	(1, 0)
04m	−0.0287	18	Distress	(0, 1)
04y	0.0447	12	Health	(1, 0)
05m	0.0461	11	Health	(1, 0)
05y	−0.1001	23	Distress	(0, 1)
06m	0.0691	6	Health	(1, 0)
06y	0.0536	9	Health	(1, 0)

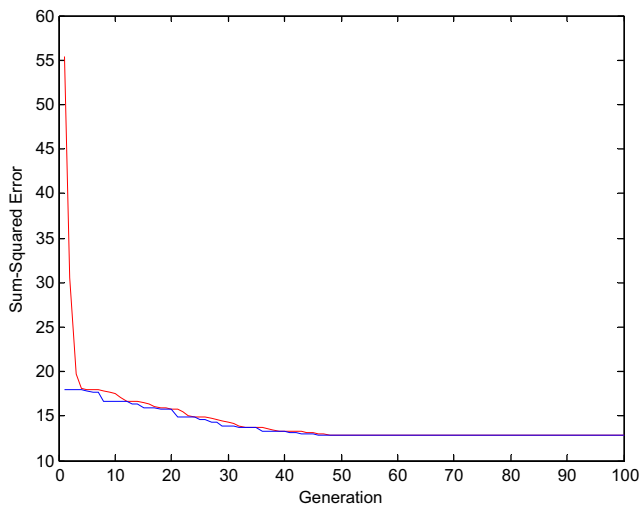


Fig. 4. Error curve of genetic algorithm.

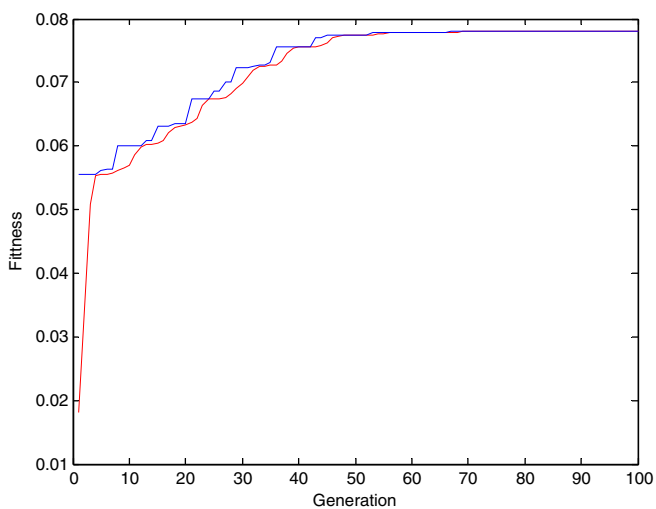


Fig. 5. Fitness curve of genetic algorithm.

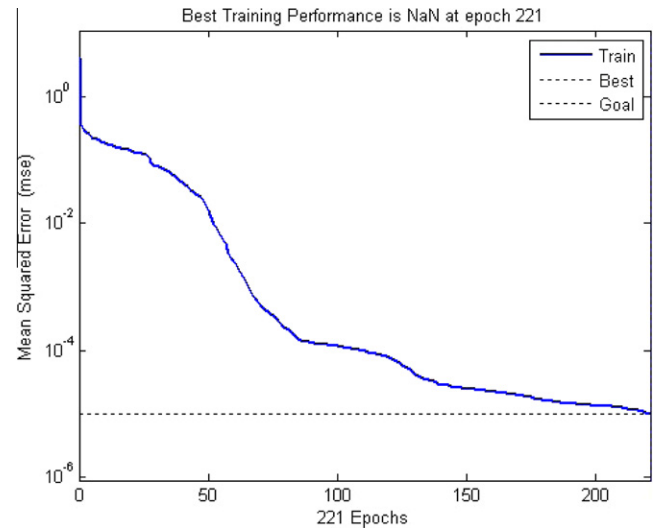


Fig. 6. Training convergence process of neural network.

Table 4
The results of the models' filtered financial indicators.

Verifying models	Model 1	V1	V3	V15	V4	V2	V10	V7	V6
	Model 2	V1	V3	V15	V2	V4	V10	V7	V6
	Model 3	V1	V15	V3	V2	V8	V10	V4	V7
	Model 4	V1	V15	V3	V2	V8	V7	V6	V16
	Model 5	V1	V15	V3	V2	V7	V6	V16	V13
	Model 6	V1	V15	V3	V2	V7	V6	V16	V13
	Model 7	V1	V3	V2	V15	V7	V6	V16	V13
	Model 8	V1	V2	V3	V7	V10	V6	V9	V11
Testing models	Model 9	V1	V2	V3	V15	V7	V9	V11	V13

option to re-construct the model with longitudinal data streams of the company since the objective of this type of method is to make prediction of financial state of a company on the base of its historical data.

5. Conclusions

From the results of this experiment, we can get the following conclusions:

- (1) With the development of an enterprise, the enterprise's financial situation is constantly changing. PCA is used in each dynamic model. Due to the dynamic inflow of financial data, the enterprise's financial situation of the same year would be scored and ranked differently. So, the discrimination results about the same year may be different in different models. That is, in a model, the financial situation at a certain year in the model i is defined as non-distress. While in another model j , the financial situation may be discriminated as distress. The reason is that the definition of enterprise financial situation always changes with the process of its life cycle. Compared with static modeling, the dynamic modeling established in this article meet the needs of financial distress prediction better in new economic environment and business operating environment with the lapse of time.
- (2) Each FDP model of an individual enterprise is trained with longitudinal data streams on the 22 half-year periods of financial indicator data to ensure the model's stability. The model is constantly and dynamically updated to predict financial situation of the next period. Accuracies of previous

Table 5

Contrast network inference results.

Prediction year	Target sample outputs	Dynamic GA-NN			Static GA-NN		
		Network inference outputs	Network warning results	Outcome evaluation	Network inference outputs	Network warning results	Outcome evaluation
2006y	(1, 0)	(1.0892, -0.0977)	Health	Correct	(1.2015, -0.0902)	Health	Correct
2007m	(0, 1)	(-0.0667, 1.1321)	Distress	Correct	(-0.0617, 1.0983)	Distress	Correct
2007y	(1, 0)	(0.9284, 0.0716)	Health	Correct	(1.1346, -0.1282)	Health	Correct
2008m	(0, 1)	(0.1176, 0.8674)	Distress	Correct	(-0.2670, 1.2635)	Distress	Correct
2008y	(0, 1)	(-0.0485, 1.0433)	Distress	Correct	(0.1849, 0.8183)	Distress	Correct
2009m	(0, 1)	(0.0254, 0.9574)	Distress	Correct	(-0.3601, 1.3651)	Distress	Correct
2009y	(0, 1)	(-0.1117, -0.9801)	Distress	Correct	(0.8761, 0.1326)	Health	Incorrect
2010m	(1, 0)	(0.9559, 0.0495)	Health	Correct	(0.8153, 0.1486)	Health	Correct

Table 6

Generalization results.

Prediction year	600606			600620			600613		
	Target sample outputs	Network inference outputs	Outcome evaluation	Target sample outputs	Network inference outputs	Outcome evaluation	Target sample outputs	Network inference outputs	Outcome evaluation
2006y	(1, 0)	(1.0892, -0.0977)	Correct	(1, 0)	(1.1008, -0.1003)	Correct	(1, 0)	(0.0143, 1.0255)	Incorrect
2007m	(0, 1)	(-0.0667, 1.1321)	Correct	(0, 1)	(0.9730, 0.0682)	Incorrect	(0, 1)	(-0.0189, 0.9967)	Correct
2007y	(1, 0)	(0.9284, 0.0716)	Correct	(1, 0)	(0.8658, 0.1176)	Correct	(0, 1)	(0.9675, 0.0268)	Incorrect
2008m	(0, 1)	(0.1176, 0.8674)	Correct	(1, 0)	(0.9795, 0.0347)	Correct	(1, 0)	(0.9910, 0.0119)	Correct
2008y	(0, 1)	(-0.0485, 1.0433)	Correct	(1, 0)	(0.8854, -0.033)	Correct	(0, 1)	(-0.1135, 1.1060)	Correct
2009m	(0, 1)	(0.0254, 0.9574)	Correct	(0, 1)	(0.1026, 0.9094)	Correct	(1, 0)	(0.0252, 1.0358)	Incorrect
2009y	(0, 1)	(-0.1117, -0.9801)	Correct	(1, 0)	(1.0065, 0.0130)	Correct	(1, 0)	(1.0118, -0.0076)	Correct
2010m	(1, 0)	(0.9559, 0.0495)	Correct	(0, 1)	(-0.0161, 0.9940)	Correct	(0, 1)	(-0.0586, 1.0937)	Correct

eight models effectively illustrate the feasibility of the medium-term prediction prior to 0.5 year which can serve the enterprise's internal business management. In addition, the model established by longitudinal data streams is more suitable for an individual company's financial distress prediction. The model's generalization ability on a company from the same industry is also acceptable.

- (3) GA-NN model uses the financial data of period $(t - 0.5)$ as inputs to predict the financial situation of period t , which is ex-ante prediction. PC model uses the financial data of period t as inputs to determinate the actual financial situation of period t , which is ex-post evaluation. The combination of ex-ante prediction and ex-post evaluation has more persuasiveness for the model's effectiveness.

This paper proposes a dynamic predictive model, which takes the outputs of PCA module in determining the enterprise's financial situation as training target labels in neural network. With the lapse of time, the sample data is continuously added and deleted in order to verify the ex-ante prediction effectiveness by ex-post evaluation. By its vertical generalized prediction ability, we ultimately carry out the dynamic FDP based on the enterprise's longitudinal data streams. In the end, the experiment shows that the dynamic model is effective and feasible. This research has the following limitations. The longitudinal sample emphasizes time series. FDP based on the dynamic modeling is more applicable to predict financial distress of the enterprises themselves. It can also be used to predict financial distress of some other companies in the same industry. However, this model cannot be used in enterprises with short history due to lack of sample data.

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